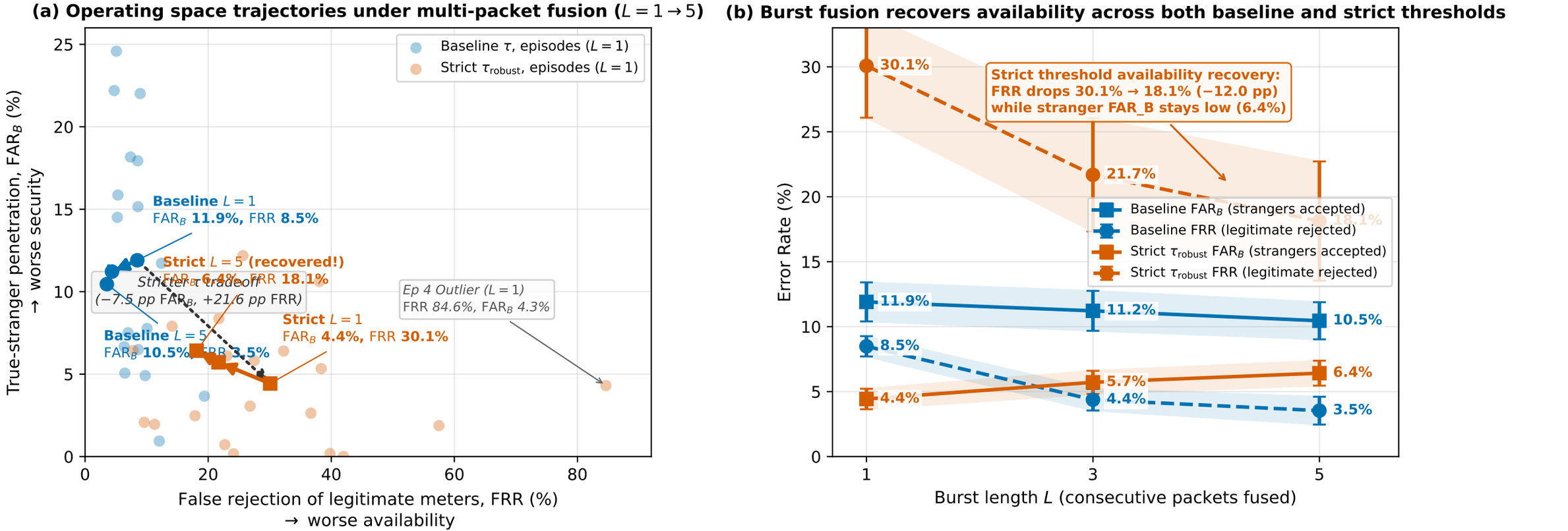


Fig. 2. Joint evaluation of decision threshold strictness and multi-packet burst fusion across 20 evaluation episodes ($N = 26{,}000$ raw transmissions). Panel (a): Operating space trajectories in (FRR, FAR_B) as burst length scales from $L = 1$ to $L = 5$. While calibrating to τ_{robust} drastically cuts stranger intrusion (11.9% → 4.4%), it inflicts a severe availability penalty (FRR : 8.5% → 30.1%). Multi-packet decision fusion ($L = 5$) pushes both operating regimes toward the ideal origin: under τ_{Youden}, FRR drops to 3.5% ($FAR_B = 10.5\%$); under strict τ_{robust}, fusion recovers 12.0 pp of lost availability (FRR : 30.1% → 18.1%) while keeping stranger penetration tightly suppressed at 6.4%. Panel (b): Solid lines denote stranger penetration (FAR_B); dashed lines denote legitimate false rejection (FRR). Shaded bands indicate ±1 standard error across the 20 evaluation episodes.